# Insurance and Data Analytics

Himchan Jeong, PhD, FSA

Simon Fraser University

Department of Applied Statistics Seminar
Gachon University

28 October, 2026

# Curriculum Vitae

| | |
|---|---|
| 2006.3 – 2012.8 | B.B.A and B.Sc in Mathematics, Seoul National University |
| 2012.7 – 2015.1 | Assistant Manager at DB Insurance |
| 2014.6 – 2019.2 | Associate of the Society of Actuaries (ASA) |
| 2014.9 – 2016.8 | M.Sc in Statistics, Seoul National University |
| 2019.3 – Present | Fellow of the Society of Actuaries (FSA) |
| 2016.8 – 2020.7 | Ph.D in Mathematics with Thesis in Actuarial Science, University of Connecticut |
| 2020.8 – 2026.8 | Assistant Professor at Simon Fraser University, Department of Statistics & Actuarial Science |
| 2026.9 – Present | Associate Professor at Simon Fraser University, Department of Statistics & Actuarial Science |

# Insurance? Insurance!

# Traditional Form of Insurance

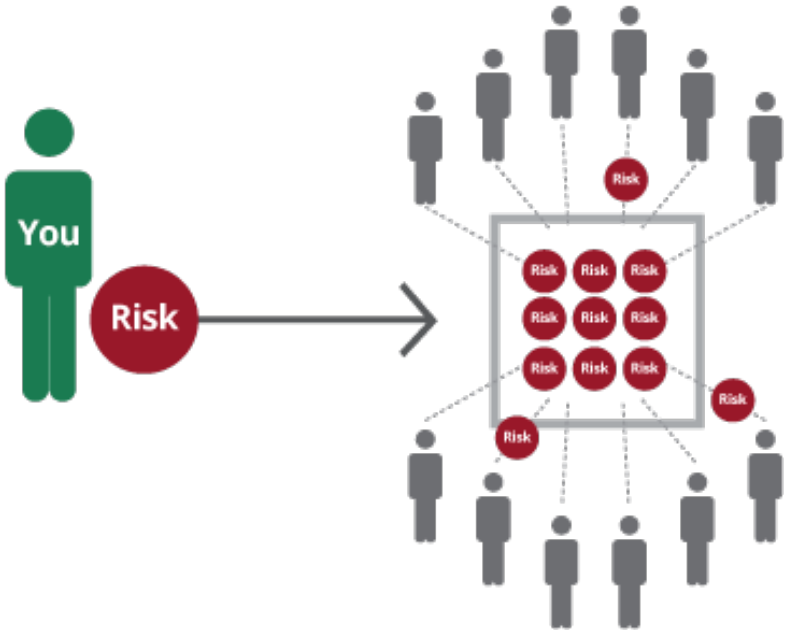


# Traditional Form of Insurance

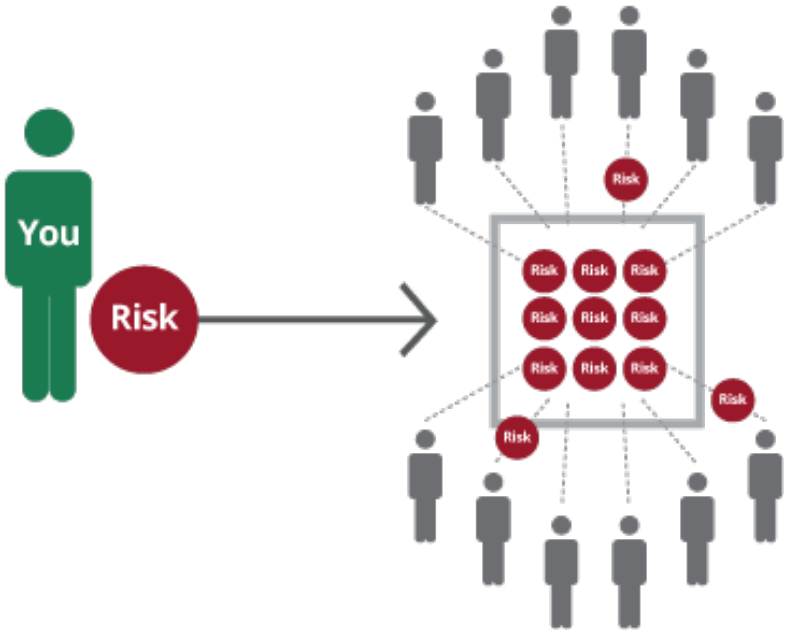


- If a seafarer was forced to throw cargo overboard to save the ship from sinking, the loss would be reimbursed collectively by his colleagues.

# Modern Form of Insurance

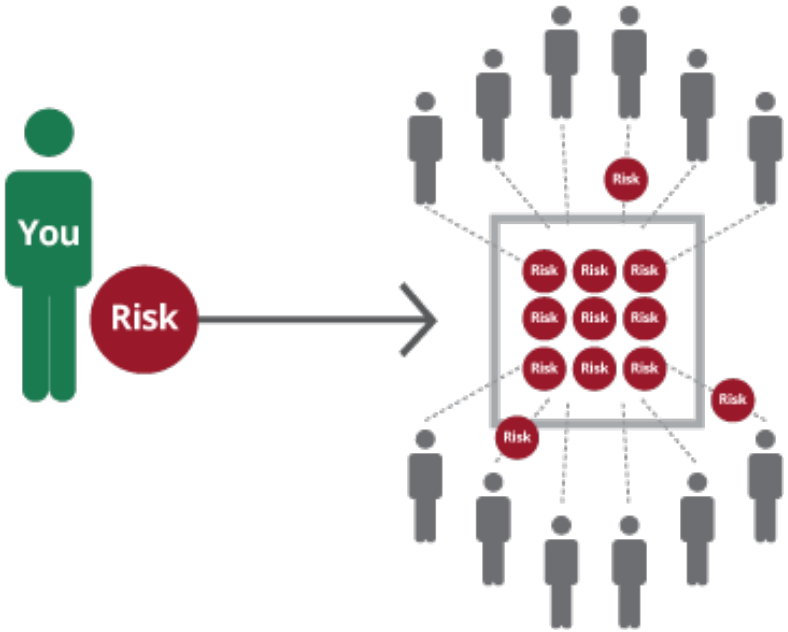


# Modern Form of Insurance




- As modern society becomes much more complicated, risks need to be shared by larger pool of people, who do not necessarily know each other.

# Modern Form of Insurance




- As modern society becomes much more complicated, risks need to be shared by larger pool of people, who do not necessarily know each other.
- As the risk pools have become anonymous, it has been critical to assess and estimate the amount of contribution to the pool.

# Example: Lung Cancer Insurance

Suppose it is known that one out of 100 people is diagnosed with lung cancer every year. If 1000 people purchase lung cancer insurance that pays $\$10,000$ to whom is diagnosed with lung cancer, how much should the annual premium per person be?

# Example: Lung Cancer Insurance

Suppose it is known that one out of 100 people is diagnosed with lung cancer every year. If 1000 people purchase lung cancer insurance that pays $\$10,000$ to whom is diagnosed with lung cancer, how much should the annual premium per person be?

- Expected total payment: $10,000 \times \frac{1{,}000}{100} = 100,000.$

# Example: Lung Cancer Insurance

Suppose it is known that one out of 100 people is diagnosed with lung cancer every year. If 1000 people purchase lung cancer insurance that pays $\$10,000$ to whom is diagnosed with lung cancer, how much should the annual premium per person be?

- Expected total payment: $10,000 \times \frac{1{,}000}{100} = 100,000.$
- Expected total payment per person: $\frac{100{,}000}{1{,}000} = 100.$

# Example: Lung Cancer Insurance

Suppose it is known that one out of 100 people is diagnosed with lung cancer every year. If 1000 people purchase lung cancer insurance that pays $\$10,000$ to whom is diagnosed with lung cancer, how much should the annual premium per person be?

- Expected total payment: $10,000 \times \frac{1{,}000}{100} = 100,000.$
- Expected total payment per person: $\frac{100{,}000}{1{,}000} = 100.$

Are we all set?

# Two Major Issues with Insurance Premium

# Two Major Issues with Insurance Premium

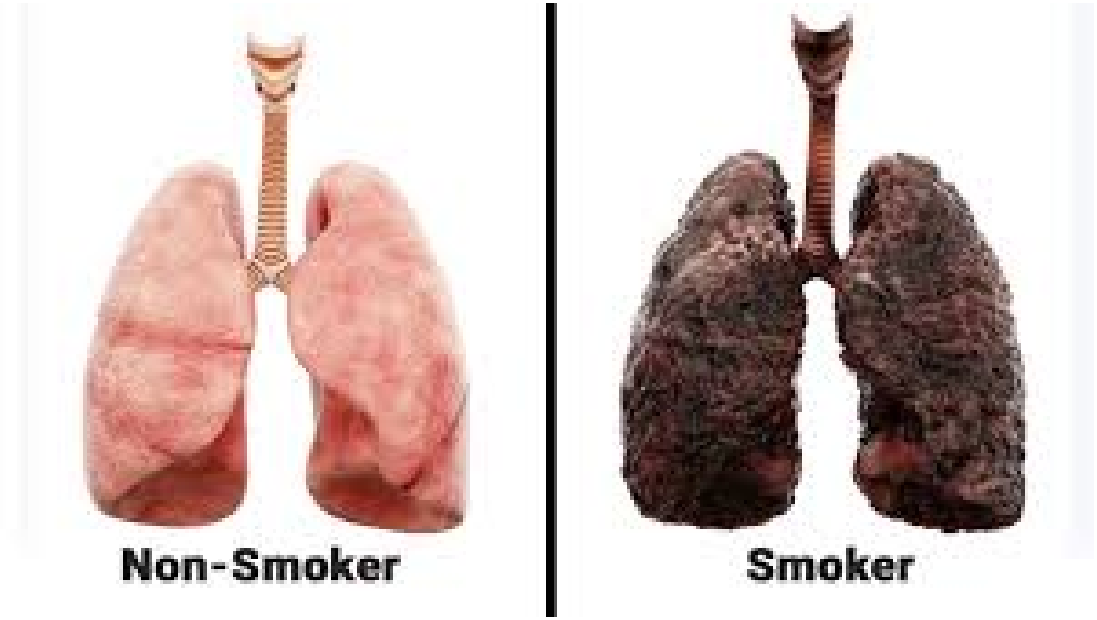
- Is the expected total payment correct?

# Two Major Issues with Insurance Premium

- Is the expected total payment correct?
- Is it fair to charge the premium evenly?

# Two Major Issues with Insurance Premium

- Is the expected total payment correct?
- Is it fair to charge the premium evenly?



# Example: Lung Cancer Insurance (Revisited)

Suppose Company A has additional information that

- 1.5 out of 100 smokers,
- 0.5 out of 100 non-smokers,

are diagnosed with lung cancer every year. Further, 500 smokers and 500 non-smokers purchase lung cancer insurance that pays $\$10,000$ to whom is diagnosed with lung cancer.

# Example: Lung Cancer Insurance (Revisited)

Suppose Company A has additional information that

- 1.5 out of 100 smokers,
- 0.5 out of 100 non-smokers,

are diagnosed with lung cancer every year. Further, 500 smokers and 500 non-smokers purchase lung cancer insurance that pays $\$10,000$ to whom is diagnosed with lung cancer.

- Expected total payment: $10,000 \times \frac{1.5 \times 500 + 0.5 \times 500}{100} = 100,000$,

# Example: Lung Cancer Insurance (Revisited)

Suppose Company A has additional information that

- 1.5 out of 100 smokers,
- 0.5 out of 100 non-smokers,

are diagnosed with lung cancer every year. Further, 500 smokers and 500 non-smokers purchase lung cancer insurance that pays $\$10,000$ to whom is diagnosed with lung cancer.

- Expected total payment: $10,000 \times \frac{1.5 \times 500 + 0.5 \times 500}{100} = 100,000$,
- Fair premium for a smoker:

# Example: Lung Cancer Insurance (Revisited)

Suppose Company A has additional information that

- 1.5 out of 100 smokers,
- 0.5 out of 100 non-smokers,

are diagnosed with lung cancer every year. Further, 500 smokers and 500 non-smokers purchase lung cancer insurance that pays $\$10,000$ to whom is diagnosed with lung cancer.

- Expected total payment: $10,000 \times \frac{1.5 \times 500 + 0.5 \times 500}{100} = 100,000$,
- Fair premium for a smoker: $10000 \times \frac{1.5}{100} = 150$,

# Example: Lung Cancer Insurance (Revisited)

Suppose Company A has additional information that

- 1.5 out of 100 smokers,
- 0.5 out of 100 non-smokers,

are diagnosed with lung cancer every year. Further, 500 smokers and 500 non-smokers purchase lung cancer insurance that pays $\$10,000$ to whom is diagnosed with lung cancer.

- Expected total payment: $10,000 \times \frac{1.5 \times 500 + 0.5 \times 500}{100} = 100,000$,
- Fair premium for a smoker: $10000 \times \frac{1.5}{100} = 150$,
- Fair premium for a non-smoker: $10000 \times \frac{0.5}{100} = 50$.

# Example: Lung Cancer Insurance (Revisited)

If Company A can differentiate the premium for a smoker and non-smoker,

| | # of Insured | Loss | Premium | Difference |
|---|---|---|---|---|
| Smoker | 500 | 75,000 | 75,000 | 0 |
| Non-smoker | 500 | 25,000 | 25,000 | 0 |
| Total | 1,000 | 100,000 | 100,000 | 0 |

# Example: Lung Cancer Insurance (Revisited)

If Company B cannot differentiate the premium for a smoker and non-smoker,

| | # of Insured | Loss | Premium | Difference |
|---|---|---|---|---|
| Smoker | 500 | 75,000 | 50,000 | -25,000 |
| Non-smoker | 500 | 25,000 | 50,000 | +25,000 |
| Total | 1,000 | 100,000 | 100,000 | 0 |

# Example: Lung Cancer Insurance (Revisited)

If Company B cannot differentiate the premium for a smoker and non-smoker,

| | # of Insured | Loss | Premium | Difference |
|---|---|---|---|---|
| Smoker | 500 | 75,000 | 50,000 | -25,000 |
| Non-smoker | 500 | 25,000 | 50,000 | +25,000 |
| Total | 1,000 | 100,000 | 100,000 | 0 |

After a year...

# Example: Lung Cancer Insurance (Revisited)

If Company B cannot differentiate the premium for a smoker and non-smoker,

| | # of Insured | Loss | Premium | Difference |
|---|---|---|---|---|
| Smoker | 500 | 75,000 | 50,000 | -25,000 |
| Non-smoker | 500 | 25,000 | 50,000 | +25,000 |
| Total | 1,000 | 100,000 | 100,000 | 0 |

After a year...

- 50 non-smokers left Company B to avoid higher premium (100 $\rightarrow$ 50),

# Example: Lung Cancer Insurance (Revisited)

If Company B cannot differentiate the premium for a smoker and non-smoker,

| | # of Insured | Loss | Premium | Difference |
|---|---|---|---|---|
| Smoker | 500 | 75,000 | 50,000 | -25,000 |
| Non-smoker | 500 | 25,000 | 50,000 | +25,000 |
| Total | 1,000 | 100,000 | 100,000 | 0 |

After a year...

- 50 non-smokers left Company B to avoid higher premium ($100 \rightarrow 50$),
- 50 smokers joined Company B to enjoy lower premium ($150 \rightarrow 100$).

# Example: Lung Cancer Insurance (Revisited)

Because of the change in the portfolio, Company B suffers $5,000$ of loss.

| | # of Insured | Loss | Premium | Difference |
|---|---|---|---|---|
| Smoker | 550 | 82,500 | 55,000 | -27,500 |
| Non-smoker | 450 | 22,500 | 45,000 | +22,500 |
| Total | 1,000 | 105,000 | 100,000 | -5,000 |

# Example: Lung Cancer Insurance (Revisited)

Because of the change in the portfolio, Company B suffers $5,000$ of loss.

| | # of Insured | Loss | Premium | Difference |
|---|---|---|---|---|
| Smoker | 550 | 82,500 | 55,000 | -27,500 |
| Non-smoker | 450 | 22,500 | 45,000 | +22,500 |
| Total | 1,000 | 105,000 | 100,000 | -5,000 |

And Company B decides to adjust the premium from 100 to 105 per person, to make up for the loss.

# Example: Lung Cancer Insurance (Revisited)

After the adjustment in premium, the portfolio looks breakeven.

| | # of Insured | Loss | Premium | Difference |
|---|---|---|---|---|
| Smoker | 550 | 82,500 | 57,750 | -24,750 |
| Non-smoker | 450 | 22,500 | 47,250 | +24,750 |
| Total | 1,000 | 105,000 | 105,000 | 0 |

# Example: Lung Cancer Insurance (Revisited)

After the adjustment in premium, the portfolio looks breakeven.

| | # of Insured | Loss | Premium | Difference |
|---|---|---|---|---|
| Smoker | 550 | 82,500 | 57,750 | -24,750 |
| Non-smoker | 450 | 22,500 | 47,250 | +24,750 |
| Total | 1,000 | 105,000 | 105,000 | 0 |

However, after another year...

# Example: Lung Cancer Insurance (Revisited)

After the adjustment in premium, the portfolio looks breakeven.

| | # of Insured | Loss | Premium | Difference |
|---|---|---|---|---|
| Smoker | 550 | 82,500 | 57,750 | -24,750 |
| Non-smoker | 450 | 22,500 | 47,250 | +24,750 |
| Total | 1,000 | 105,000 | 105,000 | 0 |

However, after another year...

- 50 non-smokers left Company B to avoid higher premium (105 $\rightarrow$ 50),

# Example: Lung Cancer Insurance (Revisited)

After the adjustment in premium, the portfolio looks breakeven.

| | # of Insured | Loss | Premium | Difference |
|---|---|---|---|---|
| Smoker | 550 | 82,500 | 57,750 | -24,750 |
| Non-smoker | 450 | 22,500 | 47,250 | +24,750 |
| Total | 1,000 | 105,000 | 105,000 | 0 |

However, after another year...

- 50 non-smokers left Company B to avoid higher premium (105 $\rightarrow$ 50),
- 50 smokers joined Company B to enjoy lower premium (150 $\rightarrow$ 105).

# Example: Lung Cancer Insurance (Revisited)

Again, because of the change in the portfolio, Company B suffers $5,000$ of loss.

| | # of Insured | Loss | Premium | Difference |
|---|---|---|---|---|
| Smoker | 600 | 90,000 | 63,000 | -27,000 |
| Non-smoker | 400 | 20,000 | 42,000 | +22,000 |
| Total | 1,000 | 110,000 | 105,000 | -5,000 |

# Example: Lung Cancer Insurance (Revisited)

Again, because of the change in the portfolio, Company B suffers $5,000$ of loss.

| | # of Insured | Loss | Premium | Difference |
|---|---|---|---|---|
| Smoker | 600 | 90,000 | 63,000 | -27,000 |
| Non-smoker | 400 | 20,000 | 42,000 | +22,000 |
| Total | 1,000 | 110,000 | 105,000 | -5,000 |

Would Company B survive in the end...?

# Insurance Data Analytics

# Data Description

Suppose we have a dataset of 1,000 observations described as follows:

# Data Description

Suppose we have a dataset of 1,000 observations described as follows:

$$
\begin{aligned}
Y: &\quad \text{Loss amount} \\
X_1: &\quad \text{Gender} \quad (1 = \text{Male},\ 0 = \text{Female}) \\
X_2: &\quad \text{Age}
\end{aligned}
$$

# Data Description

Suppose we have a dataset of 1,000 observations described as follows:

$$
\begin{aligned}
Y: &\quad \text{Loss amount} \\
X_1: &\quad \text{Gender} \quad (1 = \text{Male},\ 0 = \text{Female}) \\
X_2: &\quad \text{Age}
\end{aligned}
$$

Our task is to predict the future loss amount given available information (Gender and Age in here).

# Remark: Machine Learning

Note that machine learning can be classified into

- Supervised learning: task of learning a function that maps an input to an output based on example input-output pairs
- Unsupervised learning: type of algorithm that learns patterns from untagged data

In that sense, our task is one example of 'supervised learning'.

# Model Selection

- To predict the future loss, we need a model built upon observed data.
- No model can capture reality exactly.
- Model is nothing but an approximation of reality with some assumptions.
- Some models help us to understand the complex dynamics of reality and predict the future outcomes.

# Model Selection

- To predict the future loss, we need a model built upon observed data.
- No model can capture reality exactly.
- Model is nothing but an approximation of reality with some assumptions.
- Some models help us to understand the complex dynamics of reality and predict the future outcomes.

*"All models are wrong, but some models are useful." George Box*

# Model vs Reality



REALITY




MODEL

# Preliminary Data Analysis

**summary(Y)**

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|---|---|---|---|---|---|
| 2961 | 3337 | 3460 | 3454 | 3569 | 3901 |

# Preliminary Data Analysis

**summary(Y)**

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|---|---|---|---|---|---|
| 2961 | 3337 | 3460 | 3454 | 3569 | 3901 |

Based on this data, we may propose our first model:

# Preliminary Data Analysis

**summary(Y)**

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|---|---|---|---|---|---|
| 2961 | 3337 | 3460 | 3454 | 3569 | 3901 |

Based on this data, we may propose our first model:

$$\text{Model 1} : P = 3454 \text{ for everyone.}$$

# Preliminary Data Analysis

**summary(Y)**

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|---|---|---|---|---|---|
| 2961 | 3337 | 3460 | 3454 | 3569 | 3901 |

Based on this data, we may propose our first model:

$$\text{Model 1} : P = 3454 \text{ for everyone.}$$

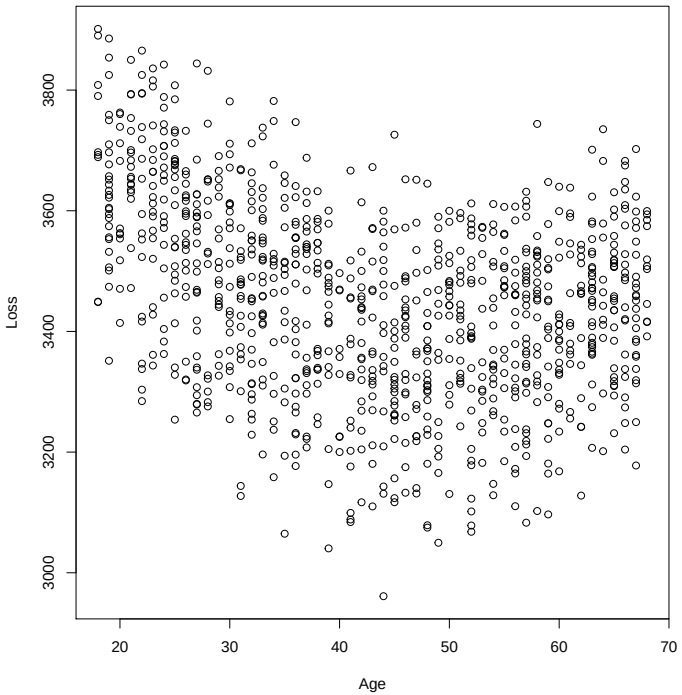
Are you happy with it?

# Preliminary Data Analysis

We might need a more refined model with risk classification.

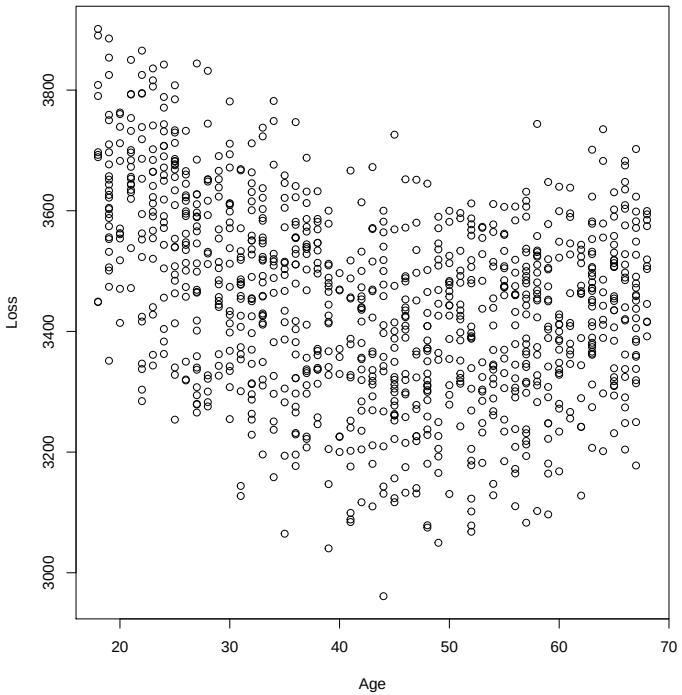
# Preliminary Data Analysis

We might need a more refined model with risk classification.



# Preliminary Data Analysis

We might need a more refined model with risk classification.



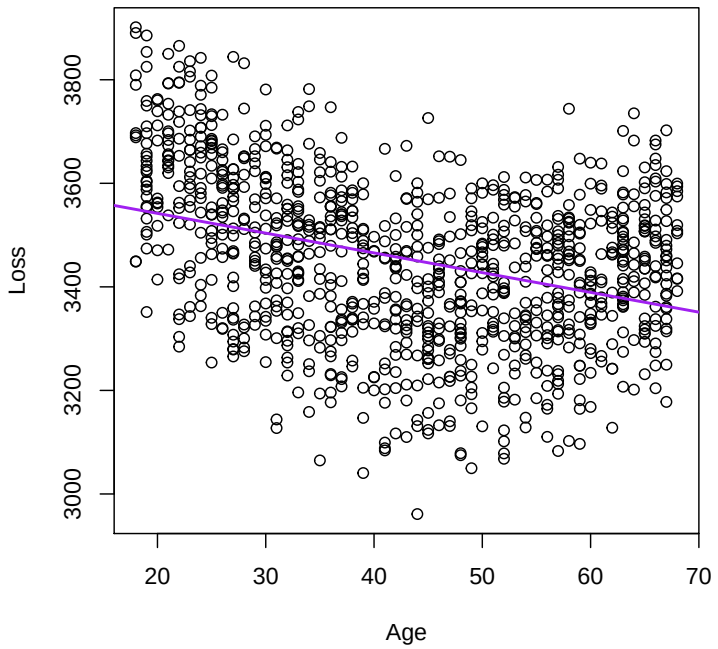

One can see that the loss amounts tend to decrease as a function of Age.

# Effect of Age: Linear

We may find the best line that captures linear effect of Age in relation to the loss amounts.

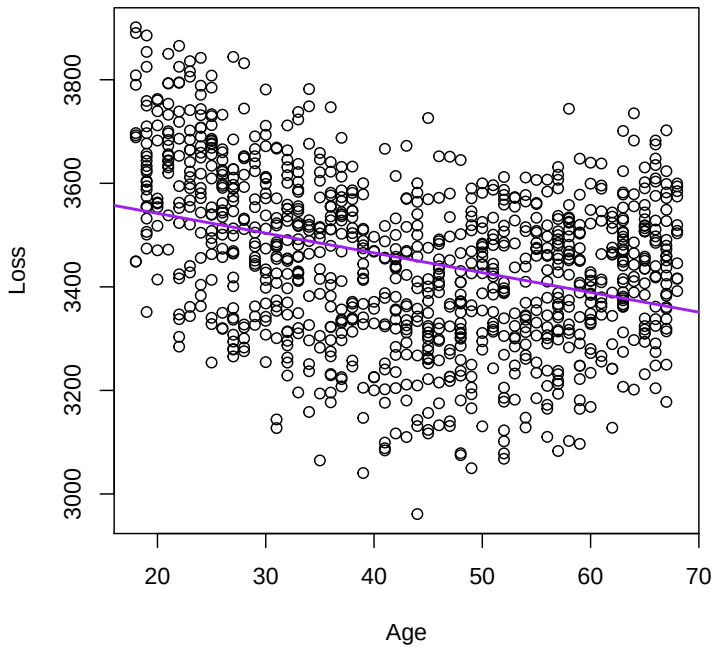
# Effect of Age: Linear

We may find the best line that captures linear effect of Age in relation to the loss amounts.



# Effect of Age: Linear

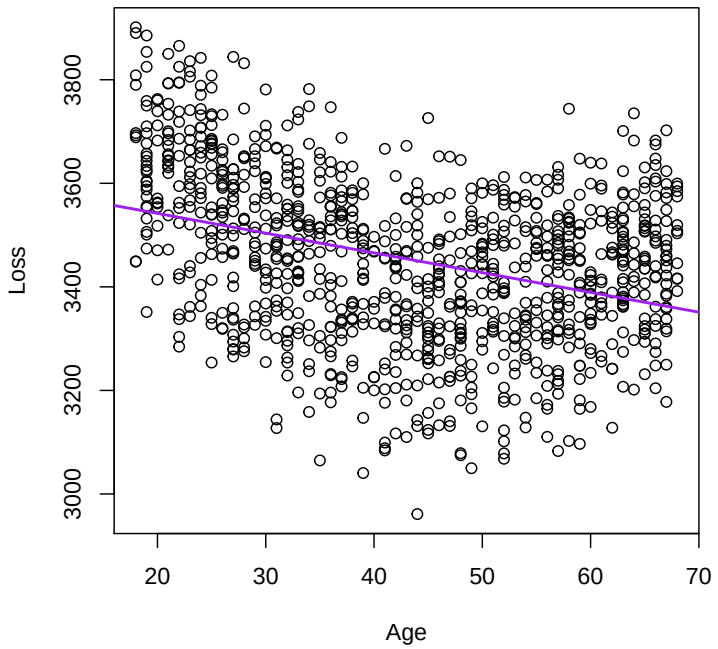
We may find the best line that captures linear effect of Age in relation to the loss amounts.




Model 2 : $P = 3618.204 - 3.815X_2$

# Effect of Age: Linear

We may find the best line that captures linear effect of Age in relation to the loss amounts.




Model 2 : $P = 3618.204 - 3.815X_2$

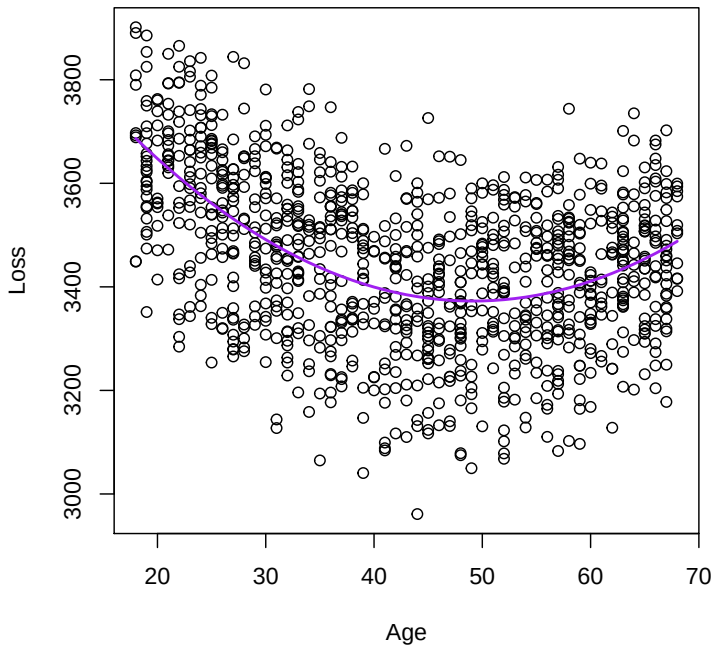
Are you happy with it?

# Effect of Age: Higher-order

We may consider adding higher-order effect of Age in relation to the loss amounts.

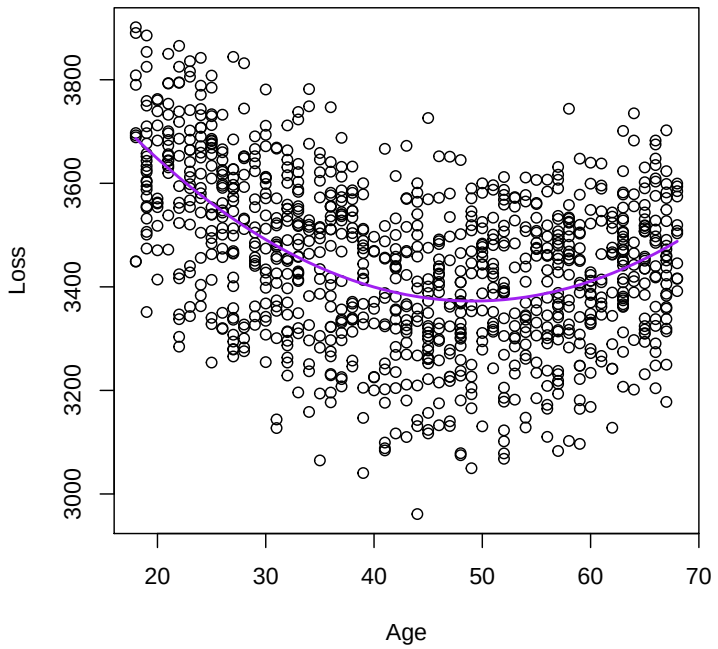
# Effect of Age: Higher-order

We may consider adding higher-order effect of Age in relation to the loss amounts.



# Effect of Age: Higher-order

We may consider adding higher-order effect of Age in relation to the loss amounts.

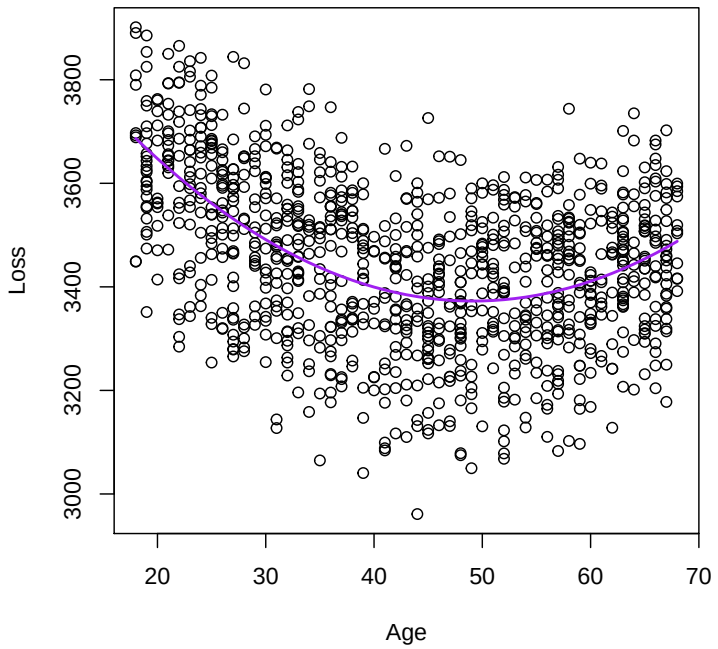



Model 3 :

$$P = 4154.728 - 31.838X_2 + 0.324X_2^2$$

# Effect of Age: Higher-order

We may consider adding higher-order effect of Age in relation to the loss amounts.



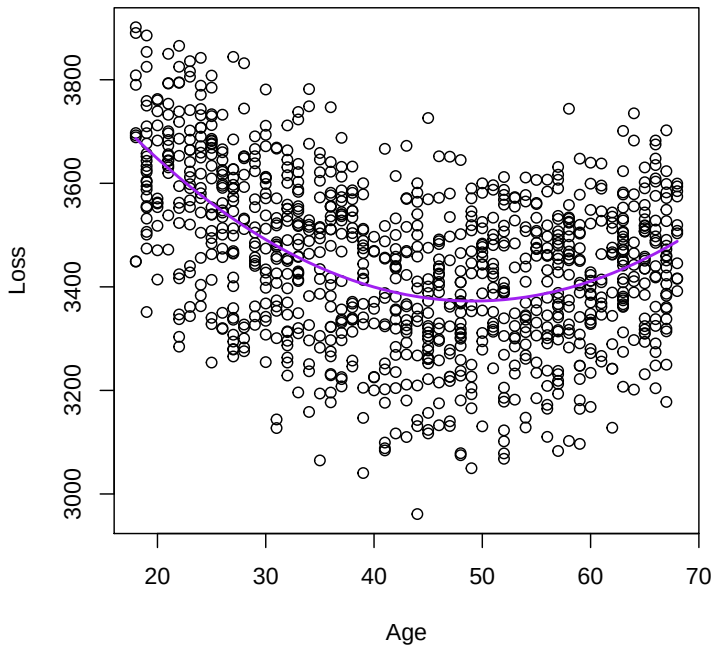

Model 3 :

$$P = 4154.728 - 31.838X_2 + 0.324X_2^2$$

Looks better :)

# Effect of Age: Higher-order

We may consider adding higher-order effect of Age in relation to the loss amounts.




Model 3 :

$$P = 4154.728 - 31.838X_2 + 0.324X_2^2$$

Looks better :)

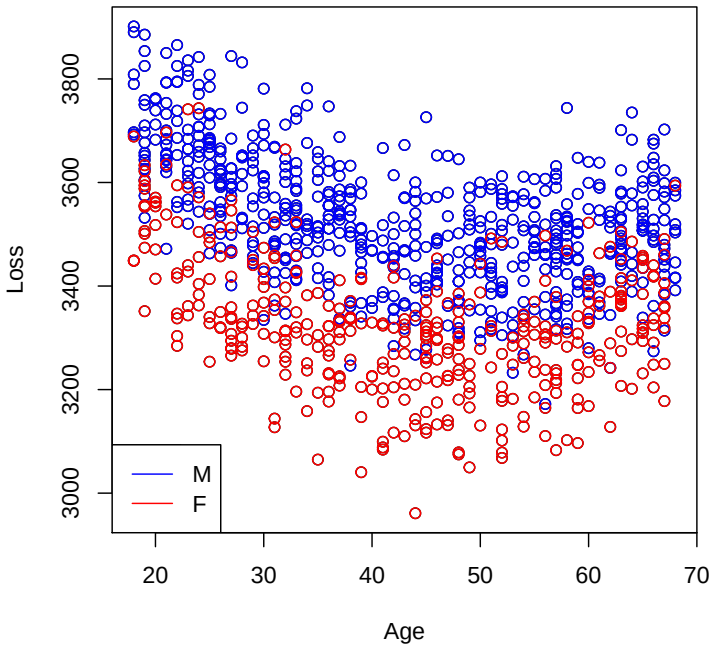
What about Gender?

# More Preliminary Data Analysis

We might plot the loss amounts for each gender with different colors.

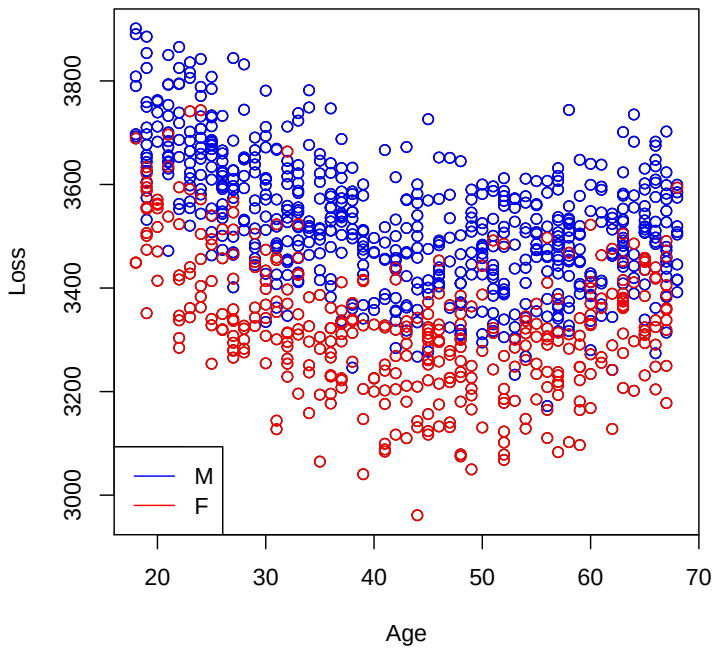
# More Preliminary Data Analysis

We might plot the loss amounts for each gender with different colors.



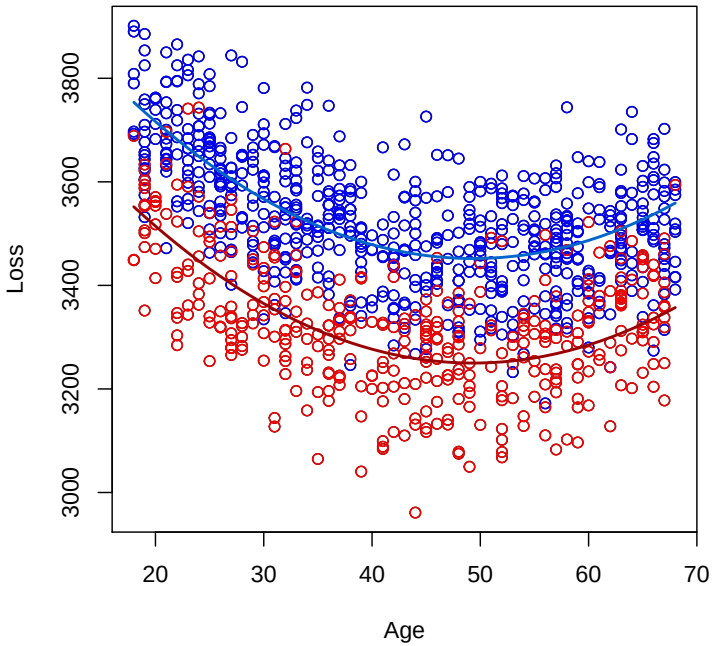
# More Preliminary Data Analysis

We might plot the loss amounts for each gender with different colors.

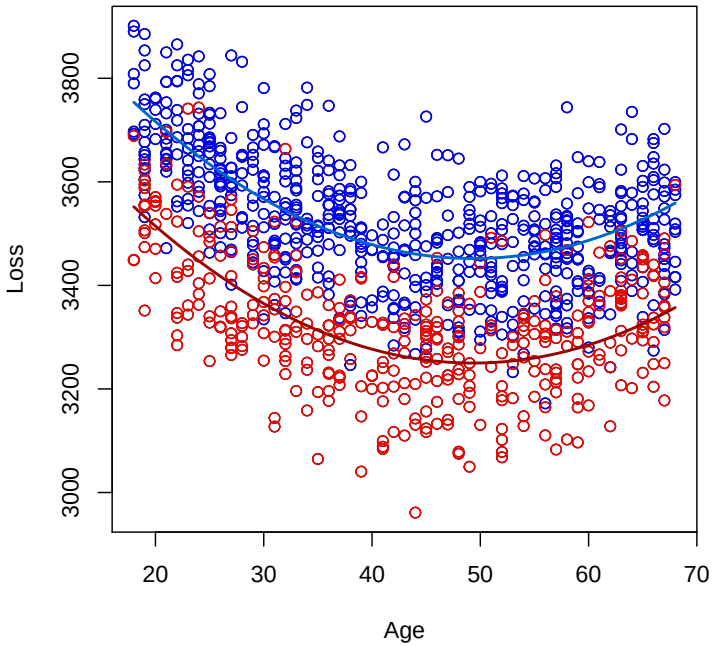



One can see that the level of loss amounts are also distinguishable with Gender.

# Putting This Together



# Putting This Together




Model 4 :

$$P = 3998.1628 - 30.3322X_2 + 0.3074X_2^2 + 201.8723X_1$$

# Validation

| | Model 1 | Model 2 | Model 3 | Model 4 |
|---|---|---|---|---|
| Difference | | | | |

# Validation

| | Model 1 | Model 2 | Model 3 | Model 4 |
|---|---|---|---|---|
| Difference | 0 | | | |

# Validation

| | Model 1 | Model 2 | Model 3 | Model 4 |
|---|---|---|---|---|
| Difference | 0 | 0 | | |

# Validation

| | Model 1 | Model 2 | Model 3 | Model 4 |
|---|---|---|---|---|
| Difference | 0 | 0 | 0 | 0 |
| Average Error | | | | |

# Validation

| | Model 1 | Model 2 | Model 3 | Model 4 |
|---|---|---|---|---|
| Difference | 0 | 0 | 0 | 0 |
| Average Error | 161.01 | | | |

# Validation

| | Model 1 | Model 2 | Model 3 | Model 4 |
|---|---|---|---|---|
| Difference | 0 | 0 | 0 | 0 |
| Average Error | 161.01 | 151.02 | | |

# Validation

| | Model 1 | Model 2 | Model 3 | Model 4 |
|---|---|---|---|---|
| Difference | 0 | 0 | 0 | 0 |
| Average Error | 161.01 | 151.02 | 138.35 | |

# Validation

| | Model 1 | Model 2 | Model 3 | Model 4 |
|---|---|---|---|---|
| Difference | 0 | 0 | 0 | 0 |
| Average Error | 161.01 | 151.02 | 138.35 | 97.94 |

# Validation

| | Model 1 | Model 2 | Model 3 | Model 4 |
|---|---|---|---|---|
| Difference | 0 | 0 | 0 | 0 |
| Average Error | 161.01 | 151.02 | 138.35 | 97.94 |

- In terms of total loss prediction, all models are acceptable.

# Validation

| | Model 1 | Model 2 | Model 3 | Model 4 |
|---|---|---|---|---|
| Difference | 0 | 0 | 0 | 0 |
| Average Error | 161.01 | 151.02 | 138.35 | 97.94 |

- In terms of total loss prediction, all models are acceptable.
- By having a better risk classification model, however, one can predict the future loss amounts more accurately.

# Flood of Data



# Flood of Data



- In this example, there were only two available inputs that can be used for risk classification, which allowed us to take (relatively simpler) step-wise approach.

# Flood of Data



- In this example, there were only two available inputs that can be used for risk classification, which allowed us to take (relatively simpler) step-wise approach.
- Due to advances in technology, nowadays we have much more data available compared to past years.

# Issues Beyond Data Analytics

- Regulations
  - Use of gender for ratemaking has been banned in EU

# Issues Beyond Data Analytics

- Regulations
  - Use of gender for ratemaking has been banned in EU
- Causality

# Issues Beyond Data Analytics

- Regulations
  - Use of gender for ratemaking has been banned in EU
- Causality
  - Shark attacks *vs* Icecream sales

# Issues Beyond Data Analytics

- Regulations
  - Use of gender for ratemaking has been banned in EU
- Causality
  - Shark attacks *vs* Icecream sales
  - Claims costs *vs* Credit scores

# Issues Beyond Data Analytics

- Regulations
  - Use of gender for ratemaking has been banned in EU
- Causality
  - Shark attacks *vs* Icecream sales
  - Claims costs *vs* Credit scores
- Internal/external Communication

# Actuarial Statistics

Subjects in actuarial science are closely related to statistical methods including but not limited to the following lists:

- Life insurance: Survival analysis, Financial time series
- Property and casualty (P&C) insurance: Categorical data analysis, Longitudinal data analysis, Multivariate analysis
- Risk management: Extreme value theory, Risk measures

# Advanced topic: Tweedie multivariate semi-parametric credibility with the exchangeable correlation

# Characteristics of Auto Insurance

- Non-normal responses:
  - Frequency: $N \in \mathbb{N} \cup \{0\}$,
  - Severity: $Y \in (0, \infty)$,
  - Compound losses: $S = Y_1 + \ldots + Y_N \in [0, \infty)$.

# Characteristics of Auto Insurance

- Non-normal responses:
  - Frequency: $N \in \mathbb{N} \cup \{0\}$,
  - Severity: $Y \in (0, \infty)$,
  - Compound losses: $S = Y_1 + \ldots + Y_N \in [0, \infty)$.
- A few covariates, (low-dimensional) $\mathbf{x}$, are available for risk classification while millions of heterogeneous policyholders are observed in a portfolio.

# Characteristics of Auto Insurance

- Non-normal responses:
  - Frequency: $N \in \mathbb{N} \cup \{0\}$,
  - Severity: $Y \in (0, \infty)$,
  - Compound losses: $S = Y_1 + \ldots + Y_N \in [0, \infty)$.
- A few covariates, (low-dimensional) $\mathbf{x}$, are available for risk classification while millions of heterogeneous policyholders are observed in a portfolio.
- There are various types of coverage within the same insurance policy.

# Characteristics of Auto Insurance

- Non-normal responses:
  - Frequency: $N \in \mathbb{N} \cup \{0\}$,
  - Severity: $Y \in (0, \infty)$,
  - Compound losses: $S = Y_1 + \ldots + Y_N \in [0, \infty)$.
- A few covariates, (low-dimensional) $\mathbf{x}$, are available for risk classification while millions of heterogeneous policyholders are observed in a portfolio.
- There are various types of coverage within the same insurance policy.
- The same policyholder is observed for multiple periods.

# Characteristics of Auto Insurance

- Non-normal responses:
  - Frequency: $N \in \mathbb{N} \cup \{0\}$,
  - Severity: $Y \in (0, \infty)$,
  - Compound losses: $S = Y_1 + \ldots + Y_N \in [0, \infty)$.
- A few covariates, (low-dimensional) $\mathbf{x}$, are available for risk classification while millions of heterogeneous policyholders are observed in a portfolio.
- There are various types of coverage within the same insurance policy.
- The same policyholder is observed for multiple periods.

$\Rightarrow$ Could be appropriate with Tweedie GLMs with multivariate random effects :)

# Notation and Data Structure

We use the following indices;

- index for policyholders ($i = 1, \ldots, I$),
- index for periods of observations for a specific policyholder $i$ ($t = 1, \ldots, T_i$), and
- index for the different types of coverages ($j = 1, \ldots, J$).

With the indicies, we denote

- multivariate claim at time $t$ of policyholder $i$: $\mathbf{Z}_{it} = (Z_{it}^{(1)}, \ldots, Z_{it}^{(J)})$ for $i = 1, \ldots, I$ and $t = 1, \ldots, T_i$, and
- policy characteristics at time $t$ of policyholder $i$: $\mathbf{x}_{it}$ for $i = 1, \ldots, I$ and $t = 1, \ldots, T_i$.

# Multivariate Tweedie Distribution with Latent Variables

the observed insurance claim, $Z_{it}^{(j)}$,

$$\mathbb{E}[Z_{it}^{(j)}|\theta_i^{(j)}] = \eta_{itj} \cdot \theta_i^{(j)}$$

- The latent variable $\theta_i^{(j)}$ captures **unobserved heterogeneity** for $(Z_{i1}^{(j)}, \ldots, Z_{iT_i}^{(j)})$ of policyholder $i$.
- $\eta_{itj} = \exp(\mathbf{x}_{it}\gamma^{(j)})$ captures **observable heterogeneity** on $j^{th}$ coverage depending on the policy characteristics of policyholder $i$ at time $t$.
-

# Tweedie distributions I

- Tweedie distributions are members of the so-called exponential dispersion family.
- They have three parameters; $\mu$, $\phi$, and $p$.
- If $Z \sim \mathcal{T}w(\mu, \phi, p)$, then we have the following properties:

$$E[Z] = \mu, \quad Var(Z) = \mu^p \cdot \phi.$$

# Tweedie distributions II

Note that Tweedie distribution contains many well-known distributions as its special cases:

- $p = 0$: Normal distribution,
- $p = 1$: Poisson distribution (when $\phi = 1$),
- $p = 2$: Gamma distribution,
- $p = 3$: Inverse Gaussian distribution,
- $p \in (1, 2)$: Compound Poisson-gamma distribution so that

$$Z \overset{d}{=} \sum_{i=1}^{N} X_i$$

where $N \sim \mathcal{P}(\lambda)$, $X_i \sim \mathcal{G}(\alpha, \gamma)$, and

$$\lambda = \frac{\mu^{2-p}}{\phi(2-p)}, \ \alpha = \frac{2-p}{p-1}, \ \gamma = \phi(p-1)\mu^{p-1}.$$

# Multivariate Tweedie Distribution with Latent Variables

Let us assume that the observed insurance claim, $Z_{it}^{(j)}$, follows a Tweedie distribution given the latent variable $\theta_i^{(j)}$ as follows:

$$Z_{it}^{(j)} | \theta_i^{(j)} \sim \mathcal{T}w(\eta_{itj}\theta_i^{(j)}, \frac{\phi_j}{\psi_{itj}}, p_j),$$

where $\mathbb{E}[Z_{it}^{(j)} | \theta_i^{(j)}] = \eta_{itj} \cdot \theta_i^{(j)}$ and $\mathbb{V}ar[Z_{it}^{(j)} | \theta_i^{(j)}] = (\eta_{itj}\theta_i^{(j)})^{p_j} \frac{\phi_j}{\psi_{itj}}$.

- The latent variable $\theta_i^{(j)}$ captures **unobserved heterogeneity** for $(Z_{i1}^{(j)}, \ldots, Z_{iT_i}^{(j)})$ of policyholder $i$.
- $\eta_{itj} = \exp(\mathbf{x}_{it}\gamma^{(j)})$ captures **observable heterogeneity** on $j^{th}$ coverage depending on the policy characteristics of policyholder $i$ at time $t$.
- $\psi_{itj}$ is the exposure for $j^{th}$ coverage of policyholder $i$ at time $t$, which is known in advance.

# Longitudinal data with multivariate responses

By using the Tweedie family, one can handle the non-normalness of the response variables (insurance claims). But how about the following properties?

- Multiple insurance coverage (multivariate responses)
- Repeated observations of the same policyholder over time (longitudinality)

We use to handle such properties.

# Longitudinal data with multivariate responses

By using the Tweedie family, one can handle the non-normalness of the response variables (insurance claims). But how about the following properties?

- Multiple insurance coverage (multivariate responses)
- Repeated observations of the same policyholder over time (longitudinality)

We use correlated to handle such properties.

# Longitudinal data with multivariate responses

By using the Tweedie family, one can handle the non-normalness of the response variables (insurance claims). But how about the following properties?

- Multiple insurance coverage (multivariate responses)
- Repeated observations of the same policyholder over time (longitudinality)

We use correlated latent factors (or random effects) to handle such properties.

# Why Random Effects?

The following hypothetical example shows us that we might capture the unobserved heterogeneity in risk by observing the residual after controlling for the effects of observed covariates, which can be explained in terms of random effects for policyholders A and B.

| Year | Gender | Age | Vehicle Size | Policyholder A | | Policyholder B | |
|---|---|---|---|---|---|---|---|
| | | | | # of Claim | Amt of Claim | # of Claim | Amt of Claim |
| 2015 | M | 45 | Medium | 0 | 0 | 1 | 500 |
| 2016 | M | 46 | Medium | 0 | 0 | 2 | 4000 |
| 2017 | M | 47 | Large | 1 | 200 | 1 | 8000 |

Note that random effects model has a good connection with credibility theory as well.

# (Correlated) Latent Factors

- The latent factors $\boldsymbol{\theta}_i = (\theta_i^{(1)}, \ldots, \theta_i^{(J)})$ account for the unobservable heterogeneity of risks on the claims of policyholder $i$ over time.
- One need to assume (without specification of the full distribution)

$$\mathbb{E}[|\theta_i^{(j)}|^{\max(p_j,2)}] < \infty \text{ and } \mathbb{E}[\theta_i^{(j)}] = 1 \text{ for all } j = 1, \ldots, \; i = 1, \ldots, I, \tag{1}$$

to assure well-definedness of the credibility premium and identifiability issues, respectively. Further, we denote

$$\mathbb{V}ar[\theta_i^{(j)}] = \sigma_j^2 \text{ and } \mathrm{Corr}(\theta_i^{(j)}, \theta_i^{(j')}) = \rho_{j,j'}. \tag{2}$$

# Usual Auto Insurance Coverage - Liabilities

- BI (Bodily Injuries): Covers medical expenses, lost wages, and legal fees if you cause an accident that injures or kills another person. It does not cover your own injuries.
- PD (Property Damages): Pays for damages you cause to someone else's property, such as their vehicle, fence, or building. It does not cover damages to your own car.

# Usual Auto Insurance Coverage - Non-liabilities

- Comprehensive: Covers damages to your own car caused by non-collision events, such as theft, vandalism, natural disasters, falling objects, and animal collisions.
- Collision: Covers damages to your own car caused by collision events.

# Usual Auto Insurance Coverage - Non-liabilities

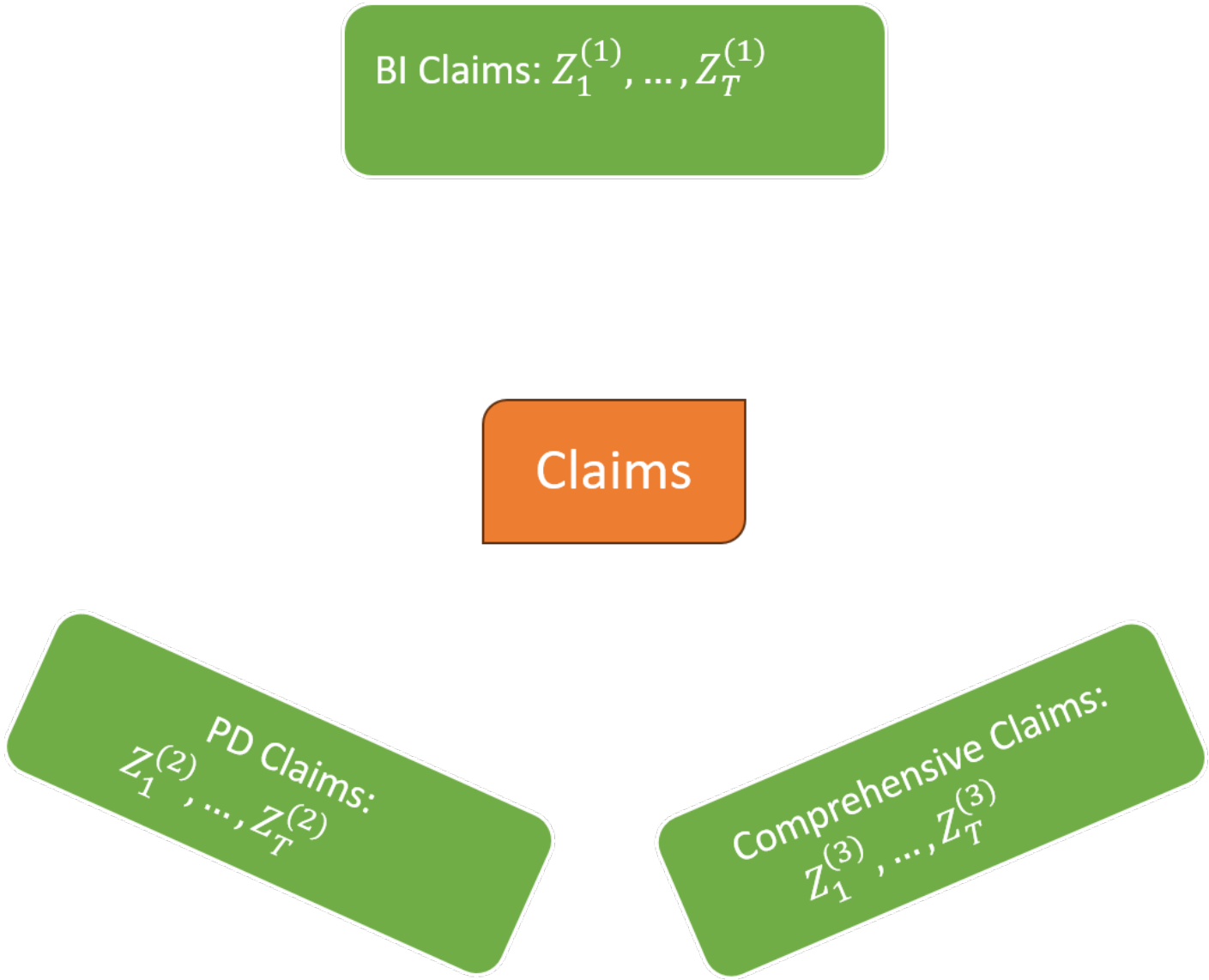
- Comprehensive: Covers damages to your own car caused by non-collision events, such as theft, vandalism, natural disasters, falling objects, and animal collisions.
- Collision: Covers damages to your own car caused by collision events.



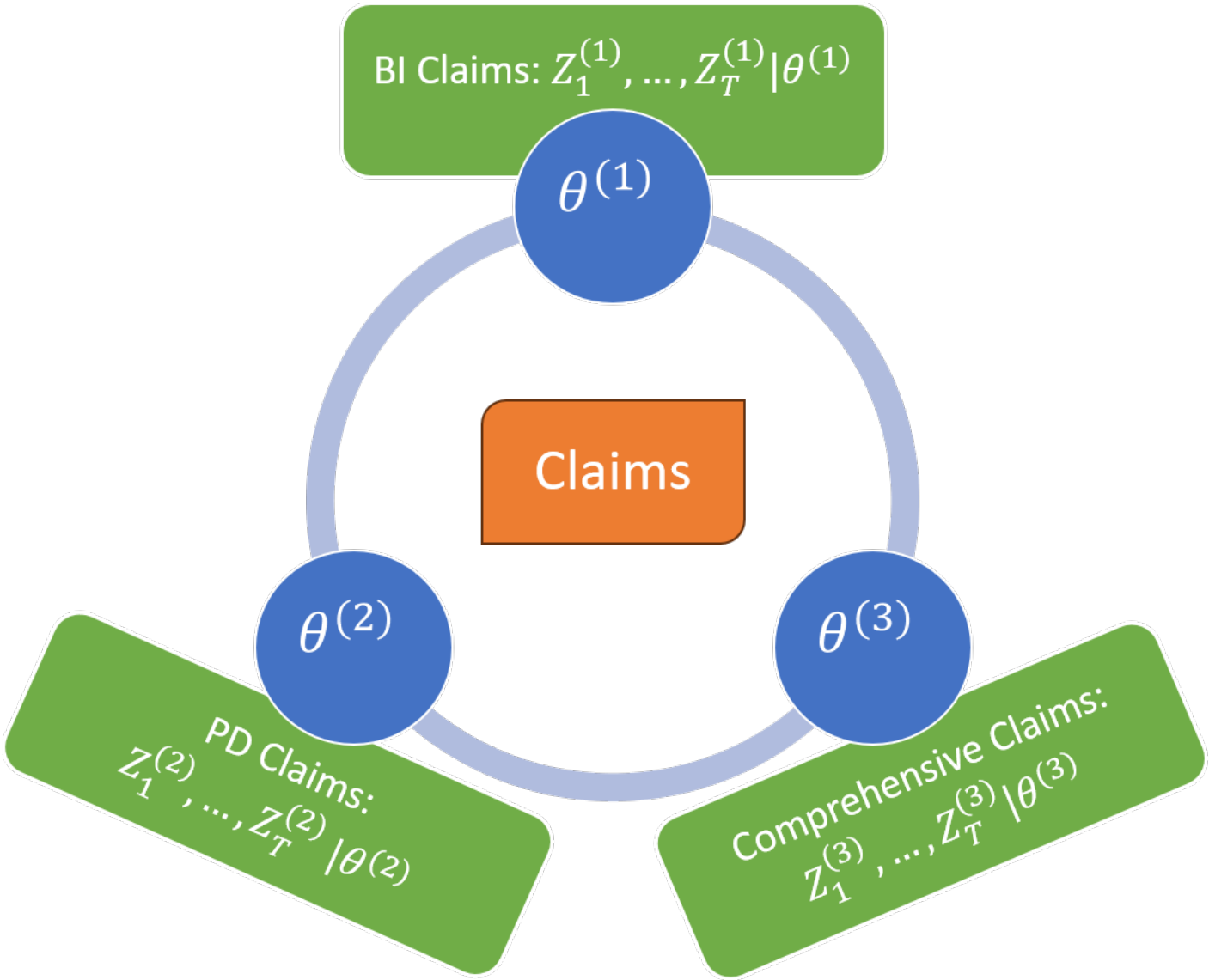

What kind of a truck/SUV can withstand hitting a moose and function afterwards? - Quora

Visit ›

# Multivariate Hierarchical Claims Models



# Multivariate Hierarchical Claims Models



# Bonus-malus Factor

Recall that we assumed $\mathbb{E}[Z_{it}^{(j)}|\theta_i^{(j)}] = \eta_{itj} \cdot \theta_i^{(j)}$, where

- $\eta_{itj} = \mathbb{E}[Z_{it}^{(j)}] = \exp(\mathbf{x}_{it}\gamma^{(j)})$ captures **observable heterogeneity** on $j^{th}$ coverage depending on the policy characteristics of policyholder $i$ at time $t$.
  - Different people with the same $\exp(\mathbf{x}_{it}\gamma^{(j)})$ would have the same $\eta_{itj}$.
- The latent variable $\theta_i^{(j)}$ captures **unobserved heterogeneity** for $(Z_{i1}^{(j)}, \ldots, Z_{iT_i}^{(j)})$ of policyholder $i$.
  - $\mathbb{E}[\theta_i^{(j)}|\mathcal{F}_T]$ could be distinct for different people (depending on the claim histories).

Our main interest is to approximate $\hat{\theta}_i^{(j)} \simeq \mathbb{E}[\theta_i^{(j)}|\mathcal{F}_T]$, which is also known as the **bonus-malus factor**.

# Bayesian Premium

After the model specifications, our primary concern is to obtain the Bayesian premium of claim from $j^{th}$ coverage at time $T+1$ given all the past actual and expected claims available up to time $T$ from the multiple coverages, which can be expressed as follows:

$$\begin{aligned}
&\mathbb{E}[Z_{T+1}^{(j)}|\mathcal{F}_T] \\
&= \int \mathbb{E}[Z_{T+1}^{(j)}|\theta^{(j)}] \cdot p\left(\theta^{(j)}|\mathcal{F}_T\right) d\theta^{(j)}, \\
&= \eta_{T+1,j} \cdot \int \theta^{(j)} p\left(\theta^{(j)}|\mathcal{F}_T\right) d\theta^{(j)},
\end{aligned}$$

where $\mathcal{F}_T = \{Z_t^{(j)}, \eta_{tj}\}_{t=1,\ldots,T,\, j=1,\ldots,J}$ and $p\left(\theta^{(j)}|\mathcal{F}_T\right)$ is the posterior density of $\theta^{(j)}$ given the past actual and expected claims available up to time $T$.

# MCMC and its Limitation

- It is usually not feasible to obtain a closed form of the posterior density of $\theta^{(j)}$ so that one might consider using MCMC to sample $\theta^{(j)}$.
- The most significant problem with MCMC is scalability in here.
  - It is not unusual to observe thousands of heterogeneous policyholders in the P&C insurance portfolio.
  - That being said, if one uses an MCMC based method to approximate the posterior mean of $\theta_i$ for $i = 1, \ldots, M$, then it is expected that the computation time with MCMC would increase proportional to $M$, which could be even more than a million in some cases.

# Bonus-malus Factor (Revisited)

Recall that we assumed $\mathbb{E}[Z_{it}^{(j)}|\theta_i^{(j)}] = \eta_{itj} \cdot \theta_i^{(j)}$, where

- $\eta_{itj} = \mathbb{E}[Z_{it}^{(j)}] = \exp(\mathbf{x}_{it}\gamma^{(j)})$ captures **observable heterogeneity** on $j^{th}$ coverage depending on the policy characteristics of policyholder $i$ at time $t$.
  - Different people with the same $\exp(\mathbf{x}_{it}\gamma^{(j)})$ would have the same $\eta_{itj}$.
- The latent variable $\theta_i^{(j)}$ captures **unobserved heterogeneity** for $(Z_{i1}^{(j)}, \ldots, Z_{iT_i}^{(j)})$ of policyholder $i$.
  - $\mathbb{E}[\theta_i^{(j)}|\mathcal{F}_T]$ could be distinct for different people (depending on the claim histories).

Our main interest is to approximate $\hat{\theta}_i^{(j)} \simeq \mathbb{E}[\theta_i^{(j)}|\mathcal{F}_T]$, which is also known as the **bonus-malus factor**.

# Aggregated Claims Experience over Time

## Lemma

*If we define*

$$Z_{\bullet}^{*(j)} = \frac{\sum_{t=1}^{T} Z_t^{(j)} \cdot w_{tj}}{\sum_{t=1}^{T} \eta_{tj} \cdot w_{tj}} \quad \textit{where} \quad w_{tj} = \eta_{tj}^{1-p_j}/\phi_{tj}, \tag{3}$$

*then*

$$\mathbb{E}[Z_{\bullet}^{*(j)}|\theta^{(j)}] = \theta^{(j)} \ \textit{and} \ \mathbb{V}ar[Z_{\bullet}^{*(j)}|\theta^{(j)}] = \frac{(\theta^{(j)})^{p_j}}{\sum_{t=1}^{T} \eta_{tj}^{2-p_j}/\phi_{tj}}.$$

Note that $Z_{\bullet}^{*(j)}$ intuitively means

$$Z_{\bullet}^{*(j)} = \frac{\text{sum of past } \textbf{actual} \text{ claims}}{\text{sum of past } \textbf{expected} \text{ claims}},$$

which implies if $Z_{\bullet}^{*(j)} > 100\%$ (resp, $< 100\%$) then the policyholder has found to be risky more (resp, less) than expected.

# Aggregated Claims Experience over Time

Thus, it is observed that $Z_{\bullet}^{*(j)}$

# Aggregated Claims Experience over Time

Thus, it is observed that $Z_{\bullet}^{*(j)}$

- serves as **an observable proxy of** (the unobserved heterogeneity) $\theta^{(j)}$,

# Aggregated Claims Experience over Time

Thus, it is observed that $Z_{\bullet}^{*(j)}$

- serves as **an observable proxy of** (the unobserved heterogeneity) $\theta^{(j)}$,
- and $\mathbb{E}[Z_{\bullet}^{*(j)}] = \mathbb{E}[\theta^{(j)}] = 1 (= 100\%)$.

# Aggregated Claims Experience over Time

Thus, it is observed that $Z_{\bullet}^{*(j)}$

- serves as **an observable proxy of** (the unobserved heterogeneity) $\theta^{(j)}$,
- and $\mathbb{E}[Z_{\bullet}^{*(j)}] = \mathbb{E}[\theta^{(j)}] = 1 (= 100\%)$.

While it is tempting to use $Z_{\bullet}^{*(j)}$ directly as an estimate of the **bonus-malus factor**, $\hat{\theta}_i^{(j)}$, it could be **too noisy** so we need some modifications on $Z_{\bullet}^{*(j)}$ to be used.

# (Multivariate) Linear Credibility Premium

In this regard, we consider the (multivariate) linear credibility premium, which is the best linear unbiased estimator (BLUE) of $\mathbb{E}[Z_{T+1}^{(j)}|\mathcal{F}_T]$ defined as follows:

$$P^B(Z_{T+1}^{(j)}|\mathcal{F}_T) := \widehat{\alpha}_0^{(j)} + \sum_{k=1}^{J}\sum_{t=1}^{T} \widehat{\alpha}_{tk}^{(j)} Z_t^{(j)} \tag{4}$$

where

$$\left(\widehat{\alpha}_0^{(j)}, \cdots, \widehat{\alpha}_{TJ}^{(j)}\right) := _{(\alpha_0,\cdots,\alpha_{TJ}^{(j)})\in\mathbb{R}^{TJ+1}} \mathbb{E}\left[\left(Z_{T+1}^{(j)} - (\alpha_0^{(j)} + \sum_{k=1}^{J}\sum_{t=1}^{T} \alpha_{tk}^{(j)} Z_t^{(j)})\right)^2\right]$$

$\widehat{\alpha}_{tk}^{(j)}$ for $t = 1, \cdots, T$ and $k = 1, \ldots, J$ is the contribution of $k^{th}$ claim observed in year $t$ to the credibility premium of $Z_{T+1}^{(j)}$.

# (Multivariate) Linear Credibility Premium

Based on the well-known procedure in Bühlmann and Gisler (2005), one can obtain the credibility coefficients and the corresponding credibility premium as follows:

$$P^{B}(Z_{T+1}^{(j)}|\mathcal{F}_T) := \eta_{T+1,j} + (\boldsymbol{Z}_{\bullet}^{*} - \mathbf{1}_J)^{\top} \left[\mathbb{V}ar(\boldsymbol{Z}_{\bullet}^{*})\right]^{-1} \mathrm{Cov}\left(\boldsymbol{Z}_{\bullet}^{*}, Z_{T+1}^{(j)}\right), \quad (5)$$

where $\boldsymbol{Z}_{\bullet}^{*} = (Z_{\bullet}^{*(1)}, \ldots, Z_{\bullet}^{*(J)})$.

# (Multivariate) Linear Credibility Premium

Based on the well-known procedure in Bühlmann and Gisler (2005), one can obtain the credibility coefficients and the corresponding credibility premium as follows:

$$P^{B}(Z_{T+1}^{(j)}|\mathcal{F}_T) := \eta_{T+1,j} + (\boldsymbol{Z}_{\bullet}^{*} - \mathbf{1}_J)^{\top} \left[\mathbb{V}ar(\boldsymbol{Z}_{\bullet}^{*})\right]^{-1} \mathrm{Cov}\left(\boldsymbol{Z}_{\bullet}^{*}, Z_{T+1}^{(j)}\right), \quad (5)$$

where $\boldsymbol{Z}_{\bullet}^{*} = (Z_{\bullet}^{*(1)}, \ldots, Z_{\bullet}^{*(J)})$.

Such a form can be in fact written in a more succinct way if we further assume the exchangeable correlation ($\rho_{j,j'} = \rho$ for all $j \neq j'$).

# Multivariate Linear Credibility Premium under Exchangeable Correlation

## Theorem

*If* $Corr(\theta^{(j)}, \theta^{(j')}) = \rho_{j,j'} = \rho$ *for all* $j \neq j'$, *then*

$$P^B(Z_{T+1}^{(j)}|\mathcal{F}_T) = \left[1 + \frac{(Z_\bullet^{*(j)} - 1)\eta_{\bullet j}^* + \xi_j \sum_{j' \neq j}^{J} \left( \frac{\sigma_j (Z_\bullet^{*(j')} - 1)\eta_{\bullet j'}^*}{\sigma_{j'}(V_{j'} + (1-\rho)\eta_{\bullet j'}^*)} \right)}{V_j + \eta_{\bullet j}^* + \rho \chi_j \xi_j}\right] \eta_{T+1,j}$$

$$= \left[\frac{V_j + Z_\bullet^{*(j)} \eta_{\bullet j}^* + \xi_j \sum_{j' \neq j}^{J} \left( \frac{(\frac{\sigma_j}{\sigma_{j'}} Z_\bullet^{*(j')} - \frac{\sigma_j}{\sigma_{j'}} + \rho)\eta_{\bullet j'}^*}{V_{j'} + (1-\rho)\eta_{\bullet j'}^*} \right)}{V_j + \eta_{\bullet j}^* + \rho \chi_j \xi_j}\right] \eta_{T+1,j}, \quad (6)$$

*where* $V_j = \mathbb{E}[(\theta^{(j)})^{p_j}]/\mathbb{V}ar[\theta^{(j)}]$, $\eta_{\bullet j}^* = \sum_{t=1}^{T} \eta_{tj}^{2-p_j}/\phi_{tj}$,

$\xi_j = \frac{\rho V_j}{1+\chi_j \rho(1-\rho)}$, *and* $\chi_j = \sum_{j' \neq j}^{J} \frac{\eta_{\bullet j'}^*}{V_{j'} + (1-\rho)\eta_{\bullet j'}^*}$.

# Multivariate Linear Credibility Premium under Exchangeable Correlation

- Here $Z_{\bullet}^{*(j)}$ defined in (3) works as a normalized residual that affects the credibility premium. As shown in (6), the credibility factor $\tilde{\theta}_i^{(j)} := P^B(Z_{T+1}^{(j)}|\mathcal{F}_T)/\eta_{T+1}^{(j)}$, is an increasing function of $Z_{\bullet}^{*(j)}$ and equals 1 if $Z_{\bullet}^{*(j)} = 1$ for all $j = 1, \ldots, J$.
- The impact of $Z_{\bullet}^{*(j')}$ (for $j \neq j'$, from a different insurance coverage) to $\tilde{\theta}_i^{(j)}$ becomes higher as $\rho$ increases here.
- Higher (respectively, lower) value of $V_j$ implies lower (respectively, higher) responsiveness of the credibility premium to the past experience.
- As an extreme case, $\tilde{\theta}_i^{(j)}$ converges to 1 as $V_j$ diverges to $\infty$ so that the past experience becomes not credible at all.

# Bivariate Linear Credibility Premium

If $J = 2$, the correlation structure of $\boldsymbol{\theta}$ is only determined by a single value, $\text{Corr}(\theta^{(1)}, \theta^{(2)}) = \rho$, so that the bivariate credibility premium can be deduced in an explicit form as follows:

## Corollary

*If $J = 2$ and $\text{Corr}(\theta^{(1)}, \theta^{(2)}) = \rho$, then for $j \neq j' \in \{1, 2\}$*

$$P^B(Z_{T+1}^{(j)} | \mathcal{F}_T) = \left[ \frac{V_j + Z_{\bullet}^{*(j)} \eta_{\bullet j}^* + \rho V_j \left( \frac{(\frac{\sigma_j}{\sigma_{j'}} Z_{\bullet}^{*(j')} - \frac{\sigma_j}{\sigma_{j'}} + \rho) \eta_{\bullet j'}^*}{V_{j'} + (1-\rho^2) \eta_{\bullet j'}^*} \right)}{V_j + \eta_{\bullet j}^* + \rho V_j \left( \frac{\rho \eta_{\bullet j'}^*}{V_{j'} + (1-\rho^2) \eta_{\bullet j'}^*} \right)} \right] \eta_{T+1,j}.$$

# Univariate Linear Credibility Premium

If the multivariate latent factors are mutually independent so that $\theta^{(j)} \perp \theta^{(j')}$ for all $j \neq j'$, then the multivariate credibility premium in (6) is reduced to the classical univariate credibility premium.

## Corollary

*If $\rho_{j,j'} = 0$ for all $j \neq j'$, then*

$$P^B(Z_{T+1}^{(j)}|\mathcal{F}_T) = P^B(Z_{T+1}^{(j)}|Z_\bullet^{*(j)}, \eta_{\bullet j}^*) = \left[\frac{V_j + Z_\bullet^{*(j)}\eta_{\bullet j}^*}{V_j + \eta_{\bullet j}^*}\right] \eta_{T+1,j}.$$

- Corollary 2 corresponds to the Tweedie credibility premium in Theorem 3.1 of Ohlsson and Johansson (2006).
- And it further reduces to the (exact) credibility premiums of Poisson and gamma claim responses by choosing $p_j = 1$ and $p_j = 2$ as in Frangos and Vrontos (2001).

# Shared Linear Credibility Premium

Lastly, if the multivariate latent factors are identical so that $\theta^{(j)} = \theta^{(j')}$ for all $j \neq j'$, then the multivariate credibility premium in (6) is reduced to the credibility premium with shared random effects as in Jeong and Dey (2024).

## Corollary

*If* $\theta^{(j)} = \theta^{(j')}$ *for* $j, j' \in \{1, \ldots, J\}$, *then we have*

$$\rho_{j,j'} = 1 \text{ for all } j, j', \ \sigma_j = \sigma,$$
$$V_j = \mathbb{E}[(\theta^{(j)})^{p_j}]/\mathbb{V}ar[\theta^{(j)}] = V \text{ for all } \ t = 1, \ldots, T, \ j = 1, \ldots, J,$$

*which lead to*

$$P^B(Z_{T+1}^{(j)} | \mathcal{F}_T) = \left[ \frac{V + \sum_{j=1}^J Z_\bullet^{*(j)} \eta_{\bullet j}^*}{V + \sum_{j=1}^J \eta_{\bullet j}^*} \right] \eta_{T+1,j}, \tag{7}$$

*because* $\xi_j = V$ *and* $\chi_j = \sum_{j' \neq j}^J \eta_{\bullet j'}^* / V$.

# Parameter Estimation

**Step 1 - Fixed effects, power, and dispersion parameters**

- For each of $j^{th}$ claim, the fixed effects coefficients $\gamma_j$ is estimated by using simple Tweedie GLMs to match $Z_{it}^{(j)}$ and $\eta_{itj} = \exp(\mathbf{x}_{it}\gamma^{(j)})$, which is still consistent as $\mathbb{E}[Z_{it}^{(j)}] = \eta_{itj}$ under the model specification.
- After that, the power parameter $p_j$ and dispersion parameter $\phi_j$ are estimated.
  - If $Z^{(j)}$ denotes claim frequency or severity, one can use $p_j = 1$ or $p_j = 2$ that corresponds to Poisson and gamma distributions, respectively.
  - In the case of the aggregate loss where $1 < p_j < 2$, one can use available optimization routines to estimate $p_j$ using the profile likelihood.

**Step 2 - Covariance matrix for the latent factors**

- Use method of moments to estimate $\sigma_j$, and $\rho_{j,j'}$ based on the estimated values of $\gamma_j$, $p_j$, and $\phi_j$ from the first step. Since

$$\mathbb{E}[(Z_{i\bullet}^{*(j)})^2] = \frac{(1+\sigma_j^2)^{\frac{p_j(p_j-1)}{2}}}{\eta_{i\bullet j}^*} + 1 + \sigma_j^2,$$

one can solve the following equation for $\hat{\sigma}_j$:

$$(1+\hat{\sigma}_j^2)\sum_{i=1}^{I}(\eta_{i\bullet j}^*)^2 + (1+\hat{\sigma}_j^2)^{\frac{p_j(p_j-1)}{2}}\sum_{i=1}^{I}\eta_{i\bullet j}^* - \sum_{i=1}^{I}(Z_{i\bullet}^{*(j)}\eta_{i\bullet j}^*)^2 = 0,$$

where $\eta_{i\bullet j}^*$ and $p_j$ are replaced with their estimates obtained from Step 1.

# Parameter Estimation

- Likewise, since

$$\mathbb{E}[Z_{i\bullet}^{*(j)} Z_{i\bullet}^{*(j')}] = 1 + \mathrm{Cov}(\theta_i^{(j)}, \theta_i^{(j')}) = 1 + \rho_{j,j'}\sigma_j\sigma_{j'},$$

  we have

$$\hat{\rho}_{j,j'} = \frac{\sum_{i=1}^{I}(Z_{i\bullet}^{*(j)} Z_{i\bullet}^{*(j')}\eta_{i\bullet j}^{*}\eta_{i\bullet j'}^{*} - \eta_{i\bullet j}^{*}\eta_{i\bullet j'}^{*})}{\hat{\sigma}_j\hat{\sigma}_{j'}\sum_{i=1}^{I}\eta_{i\bullet j}^{*}\eta_{i\bullet j'}^{*}}.$$

- If we assume the common correlation so that $\rho_{j,j'} = \rho$ for all $j \neq j'$, then a pooled estimator of $\rho$ is given as follows:

$$\hat{\rho} = \frac{\sum_{j\neq j'}\sum_{i=1}^{I}(Z_{i\bullet}^{*(j)} Z_{i\bullet}^{*(j')}\eta_{i\bullet j}^{*}\eta_{i\bullet j'}^{*} - \eta_{i\bullet j}^{*}\eta_{i\bullet j'}^{*})}{\sum_{j\neq j'}\hat{\sigma}_j\hat{\sigma}_{j'}\sum_{i=1}^{I}\eta_{i\bullet j}^{*}\eta_{i\bullet j'}^{*}}.$$

# Simulation Study Setup

We generate $\{S_{it}^{(j)}\}_{i=1,\ldots,M,\, t=1,\ldots,T+1,\, j=1,2,3}$ with the following hierarchical distributions:

$$S_{it}^{(j)}|\theta_i^{(j)} \sim \mathcal{T}w(\eta_{itj}\theta_i^{(j)}, \phi_j/\psi_{itj}, p_j),\ X_{itj}^{[1]} \sim \mathcal{N}(0,1),\ X_{itj}^{[2]} \sim \mathcal{BN}(0.6), \psi_{itj} = 1,$$

where $\boldsymbol{\theta}_i := (\theta_i^{(1)}, \ldots, \theta_i^{(J)})$ follows

$$\theta_i^{(j)} \sim \mathcal{LN}\left(-\log(1+\sigma_j^2)/2,\ \log(1+\sigma_j^2)\right), \mathrm{Corr}(\theta_i^{(j)}, \theta_i^{(j')}) = \rho_{j,j'}, \quad (8)$$

so that $\mathbb{E}[(\theta_i^{(j)})^{p_j}] = (1+\sigma_j^2)^{\frac{p_j(p_j-1)}{2}}$, and

$$\eta_{itj} = \exp(\mathbf{x}_{itj}\boldsymbol{\gamma}_j) = \exp(\gamma_{0j} + \gamma_{1j}X_{itj}^{[1]} + \gamma_{2j}X_{itj}^{[2]}),$$
$$\gamma_{01} = 3.0,\ \gamma_{11} = 1.5,\ \gamma_{21} = -0.20,\ \sigma_1 = \log(1.5),\ p_1 = 1.6,\ \phi_1 = 20,$$
$$\gamma_{02} = 2.5,\ \gamma_{12} = 1.0,\ \gamma_{22} = -0.25,\ \sigma_2 = \log(1.6),\ p_2 = 1.4,\ \phi_2 = 40,$$
$$\gamma_{03} = 3.5,\ \gamma_{13} = 0.5,\ \gamma_{23} = +0.15,\ \sigma_3 = \log(1.7),\ p_3 = 1.5,\ \phi_3 = 30,$$
$$\rho_{12} = \rho_{23} = \rho_{31} = 0.5,\ M = 5000,\ \ T = 5.$$

# Benchmarks for Simulation Study

We compare the following methods to compute the credibility premiums:

- Naive: $\hat{\mathbb{E}}[S^{(j)}_{i,T_{i+1}}|\mathcal{F}_{i,T_i}] = \hat{\eta}_{i,T_i+1,j}$ for $j = 1, 2, 3$ (no serial correlation),
- Univariate: $\hat{\mathbb{E}}[S^{(j)}_{i,T_{i+1}}|\mathcal{F}_{i,T_i}] = \left[\frac{V_j + Z^{*(j)}_{i\bullet}\hat{\eta}^*_{i\bullet j}}{V_j + \hat{\eta}^*_{i\bullet j}}\right]\hat{\eta}_{i,T_i+1,j}$ for $j = 1, 2, 3$ as in Corollary 2, which implicitly assumes $\rho_{j,j'} = 0$ for all $j \neq j'$ (no correlation among the multiple coverage),
- Shared: $\hat{\mathbb{E}}[S^{(j)}_{i,T_{i+1}}|\mathcal{F}_{i,T_i}] = \left[\frac{V + \sum_{i=1}^{3} Z^{*(j)}_{i\bullet}\hat{\eta}^*_{i\bullet j}}{V + \sum_{i=1}^{3} \hat{\eta}^*_{i\bullet j}}\right]\hat{\eta}_{i,T_i+1,j}$ for $j = 1, 2, 3$ as in Corollary 3, which implicitly assumes $\rho_{j,j'} = 1$ for all $j, j' \in \{1, 2, 3\}$ (perfect correlation among the multiple coverage),

# Benchmarks for Simulation Study

- Proposed:

$$\hat{\mathbb{E}}[S_{i,T_{i+1}}^{(j)}|\mathcal{F}_{i,T_i}] = \left[ \frac{V_j + Z_{i\bullet}^{*(j)}\hat{\eta}_{i\bullet j}^{*} + \xi_j \sum_{j' \neq j}^{3} \left( \frac{(\frac{\sigma_j}{\sigma_{j'}} Z_{i\bullet}^{*(j')} - \frac{\sigma_j}{\sigma_{j'}} + \rho)\hat{\eta}_{i\bullet j'}^{*}}{V_{j'} + (1-\rho)\hat{\eta}_{i\bullet j'}^{*}} \right)}{V_j + \hat{\eta}_{i\bullet j}^{*} + \rho\chi_{ij}\xi_j} \right] \hat{\eta}_{i,T+1,j}$$

  for $j = 1, 2, 3$ as in (6), which implicitly assumes $\rho_{j,j} = \rho$ for all $j \neq j'$,
- True: $\hat{\mathbb{E}}[S_{i,T_{i+1}}^{(j)}|\mathcal{F}_{i,T_i}] = \theta_i^{(j)}\hat{\eta}_{i,T_i+1,j}$ for $j = 1, 2, 3$, which assumes that the true values of latent factors $\theta_i^{(j)}$ are fully known.

# 100 times of Simulation Results - Estimation

| $j$ | 1 | 2 | 3 |
|---|---|---|---|
| $\hat{\gamma}_0$ | 3.0015 | 2.5011 | 3.5085 |
| | (0.0241) | (0.0302) | (0.0268) |
| $\hat{\gamma}_1$ | 1.4975 | 0.9976 | 0.4966 |
| | (0.0240) | (0.0315) | (0.0243) |
| $\hat{\gamma}_2$ | -0.2015 | -0.2558 | 0.1418 |
| | (0.0328) | (0.0415) | (0.0318) |
| $\hat{p}$ | 1.6000 | 1.4010 | 1.5020 |
| | (0.0000) | (0.0070) | (0.0098) |
| $\hat{\phi}$ | 20.5970 | 41.3030 | 31.4596 |
| | (0.1989) | (0.5360) | (0.5150) |
| $\exp(\hat{\sigma})$ | 1.4785 | 1.5586 | 1.6915 |
| | (0.0640) | (0.0936) | (0.0633) |
| $\hat{\rho}$ | 0.5699 (0.1047) | | |

# 100 times of Simulation Results - Out-of-sample Validation

| | $j$ | Naive | Univariate | Shared | Proposed | True |
|---|---|---|---|---|---|---|
| RMSE | 1 | 80.69 | 80.43 | 80.35 | 80.32 | 78.75 |
| | | (6.31) | (6.23) | (6.23) | (6.24) | (5.70) |
| | 2 | 41.58 | 41.51 | 41.50 | 41.47 | 40.85 |
| | | (2.28) | (2.25) | (2.24) | (2.24) | (2.11) |
| | 3 | 93.65 | 93.10 | 93.13 | 93.04 | 91.03 |
| | | (4.08) | (4.01) | (4.02) | (4.01) | (3.81) |
| DEV | 1 | 76772.66 | 76476.89 | 76410.11 | 76340.90 | 74065.92 |
| | | (1061.03) | (1046.52) | (1065.54) | (1047.80) | (1008.00) |
| | 2 | 98986.02 | 98495.54 | 98361.36 | 98241.01 | 94150.57 |
| | | (2385.62) | (2343.01) | (2321.43) | (2317.54) | (2106.93) |
| | 3 | 124268.08 | 122389.66 | 122454.37 | 122169.60 | 115631.20 |
| | | (3694.13) | (3502.79) | (3531.71) | (3499.31) | (3016.07) |

# Empirical Analysis: LGPIF data

| Categorical variables | Description | | Proportions | |
|---|---|---|---|---|
| TypeCity | Indicator for city entity: | Y=1 | 14.00 % | |
| TypeCounty | Indicator for county entity: | Y=1 | 5.78 % | |
| TypeMisc | Indicator for miscellaneous entity: | Y=1 | 11.04 % | |
| TypeSchool | Indicator for school entity: | Y=1 | 28.17 % | |
| TypeTown | Indicator for town entity: | Y=1 | 17.28 % | |
| TypeVillage | Indicator for village entity: | Y=1 | 23.73 % | |
| NoClaimCreditBC | No BC claim in three consecutive prior years: | Y=1 | 32.83 % | |
| NoClaimCreditIM | No IM claim in three consecutive prior years: | Y=1 | 42.10 % | |
| NoClaimCreditCN | No CN claim in three consecutive prior years: | Y=1 | 8.97 % | |
| NoClaimCreditPN | No PN claim in three consecutive prior years: | Y=1 | 10.96 % | |
| Continuous variables | | Minimum | Mean | Maximum |
| CoverageBC | Log coverage amount of BC claim in mm | 0 | 37.050 | 2444.797 |
| lnDeductBC | Log deductible amount for BC claim | 0 | 7.137 | 11.513 |
| CoverageIM | Log coverage amount of IM claim in mm | 0 | 0.8483 | 46.7493 |
| lnDeductIM | Log deductible amount for IM claim | 0 | 5.340 | 9.210 |
| CoverageCN | Log coverage amount of CN claim in mm | 0 | 0.0965 | 4.1398 |
| CoveragePN | Log coverage amount of PN claim in mm | 0 | 0.1584 | 25.6654 |

# Actual data vs. Simulated data

A major difference between the LGPIF dataset and the simulated data is two-fold.

- Firstly, while we assumed $T_i = 5$ for all policyholders in the simulated data, it is not necessarily true in an actual data.
- Secondly, while each policyholder held all coverage in every insurance year in the simulated data, it is not necessary for a policyholder to hold all coverage in a year in practice.
  - The following table shows the number and proportion of policies with each coverage.

| | BC | IM | CN | PN |
|---|---|---|---|---|
| Number | 5660 | 4622 | 1529 | 1638 |
| Proportion | 99.70% | 81.42% | 26.93% | 28.85% |

  - One can see that while BC and IM coverages were in most of the policies, past claims information on CN and PN are relatively few.

# Estimation of the Fixed Effects in LGPIF data

| | BC | | IM | | PN | | CN | |
|---|---|---|---|---|---|---|---|---|
| | Estimate | p-value | Estimate | p-value | Estimate | p-value | Estimate | p-value |
| (Intercept) | 7.184 | 0.000 | 9.104 | 0.000 | 8.745 | 0.000 | 8.323 | 0.000 |
| TypeCity | -0.065 | 0.912 | -1.100 | 0.002 | -1.952 | 0.000 | -0.996 | 0.000 |
| TypeMisc | -2.036 | 0.006 | -6.516 | 0.000 | -3.297 | 0.000 | -1.903 | 0.004 |
| TypeSchool | -0.157 | 0.775 | -3.878 | 0.000 | -3.422 | 0.000 | -2.086 | 0.000 |
| TypeTown | -1.932 | 0.006 | -3.003 | 0.000 | -2.987 | 0.000 | -2.848 | 0.000 |
| TypeVillage | -1.304 | 0.034 | -2.839 | 0.000 | -3.328 | 0.000 | -1.210 | 0.000 |
| Coverage | 0.002 | 0.001 | 0.111 | 0.000 | 0.109 | 0.014 | 0.000 | 0.000 |
| lnDeduct | 0.370 | 0.003 | -0.140 | 0.397 | -0.588 | 0.052 | 0.675 | 0.000 |
| NoClaimCredit | -0.457 | 0.166 | -0.255 | 0.303 | - | - | - | - |

# Estimation of the Random Effects Parameters in LGPIF data

| | BC | IM | PN | CN |
|---|---|---|---|---|
| $\hat{p}$ | 1.5143 | 1.6816 | 1.4224 | 1.4653 |
| $\hat{\phi}$ | 711.2547 | 160.6885 | 373.9267 | 338.2100 |
| $\hat{\sigma}$ | 1.3275 | 3.6379 | 0.8890 | 0.8289 |
| $\hat{\rho}$ | | 0.1604 | | |

# Out-of-Sample validation with LGPIF data

| | | Naive | Univariate | Shared | Proposed |
|---|---|---|---|---|---|
| RMSE | BC | 488062.8 | 163699.8 | 144028.7 | 136161.8 |
| | IM | 43996.1 | 21798.2 | 15376.6 | 15393.3 |
| | PN | 3597.2 | 3488.8 | 3659.0 | 3500.4 |
| | CN | 2916.3 | 2538.4 | 3000.4 | 2566.8 |
| DEV | BC | 129130.1 | 122865.4 | 122735.0 | 122193.2 |
| | IM | 85189.2 | 83559.7 | 82214.0 | 82752.3 |
| | PN | 63978.5 | 61734.6 | 66521.1 | 62020.5 |
| | CN | 52481.1 | 49082.1 | 50666.1 | 48994.2 |

# Actuarial Statistics for Energy Demand Research

Load forecasts predict the *mean*; grid decisions depend on **tails, uncertainty and differences across units**, which are the core subjects of actuarial science.

| **Actuarial tool** | **Energy question** | **Example** |
|---|---|---|
| **Extreme value theory** | How high can the peak go? | GPD over a threshold $\rightarrow$ 1-in-10-year peak load |
| **Risk measures** | How much reserve to hold? | $\mathrm{VaR}_\alpha$, $\mathrm{TVaR}_\alpha$ of the forecast error |
| **Credibility theory** | Unit with little history? | $\hat{\mu}_i = Z_i \bar{X}_i + (1 - Z_i)\mu$ for new loads, substations |
| **Bivariate credibility** | How often and how badly does it overload? | Mild $N_{it}$, severe $M_{it} \leq N_{it}$: binomial thinning |

## Takeaway

Forecasts $\rightarrow$ operational quantities: reserve margins, overload probabilities, return-level peaks.